

Jihwan Hong

csjihwanh@snu.ac.kr · Google Scholar · csjihwanh.com

Research Interests

- Video Understanding and Generation
- World Modeling
- 3D/4D Reconstruction and Generation

Education

Seoul National University 2025 – Present
M.S. in Artificial Intelligence

Korea University 2019 – 2025
B.S. in Computer Science and Engineering

Research Experience

Research Assistant, AIDAS Lab, Seoul National University, Korea Aug. 2025 – Present
Advisor: Prof. Jaeyoung Do

Research Intern, AIDAS Lab, Seoul National University, Korea Jan. 2025 – Aug. 2025
Advisor: Prof. Jaeyoung Do

Research Intern, MLV Lab, Korea University Jun. 2023 – Dec. 2024
Advisor: Prof. Hyunwoo J. Kim

Research Intern, AI Lab, Korea University Oct. 2020 – Jan. 2021
Advisor: Prof. Dongsuk Yook

Publications

When Predicting Nothing Beats SAM 3: Revisiting Evaluation in Video Object Segmentation 2026
40th Annual Conference on Neural Information Processing Systems (NeurIPS), Evaluations and Datasets Track
Jihwan Hong, Woohyeon Park, Jaeik Kim, Jaeyoung Do[†]

VIRST: Video-Instructed Reasoning Assistant for SpatioTemporal Segmentation 2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
Jihwan Hong, Jaeyoung Do[†]
Project · Paper · arXiv · Code

Report of the 5th PVUW Challenge: Towards More Diverse Modalities in Pixel-Level Understanding 2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)
Chang Liu *et al.*
Paper · arXiv

Preprints & Manuscripts

Dynin-Omni: Omnimodal Unified Large Diffusion Language Model <i>Under review</i> Jaeik Kim, Woojin Kim, Jihwan Hong , Yejoon Lee, Sieun Hyeon, Mintaek Lim, Yunseok Han, Dogeun Kim, Hoeun Lee, Hyungeun Kim, Jaeyoung Do [†] Demo · Code · Model · arXiv	2026
DM3SR: Benchmarking Deep Multi-View 3D Spatial Reasoning <i>Under review</i> Jusang Oh, Jihwan Hong , Jaeyoung Do [†]	2026
SoccerNet 2024 Challenges Results <i>arXiv preprint</i> Anthony Cioppa <i>et al.</i> arXiv	2024

Awards & Honors

3rd Place, 5th PVUW Challenge MeViS-Audio Track <i>CVPR 2026 Pixel-level Video Understanding in-the-Wild (PVUW) Workshop</i> 3rd Place of MeViS-Audio Track of the 5th PVUW: VIRST-Audio Jihwan Hong , Jaeyoung Do [†] arXiv	2026
2nd Place, CVPRW 2024 SoccerNet Challenge: Multi-View Foul Recognition <i>CVPR 2024 CVsports Workshop</i> VF-VARS: Leveraging Video Foundation Models for Video Assistant Referee Systems Jihwan Hong [*] , Youngseo Kim [*] , Junseok Lee, Jimin Lee, Sehyung Kim, Hyunwoo J. Kim [†] Technical Report	2024
1st Place, The 14th KSME Student Creative Design Competition <i>The Korean Society of Mechanical Engineers; Minister of Science and ICT Award, Government of South Korea</i> Development of an AI-Based Automatic Seat Belt Adjustment Device Code	2023

Academic Service

Reviewer <i>International Conference on Learning Representations (ICLR)</i>	2027
Reviewer <i>IEEE Transactions on Multimedia</i>	2026

Selected Projects

Anatomy- and Action-Conditioned Colonoscopic World Model <i>SNU Boramae Medical Center & NVIDIA HK</i> Developing a colonoscopic I2V world model conditioned on an anatomical topology map and actions.	Jul. 2026 – Present
Technical Drawing Registration Automation <i>POSCO DX</i> Developed a vision-language-model-based workflow for autonomous technical drawing registration.	Apr. 2025 – Jun. 2025